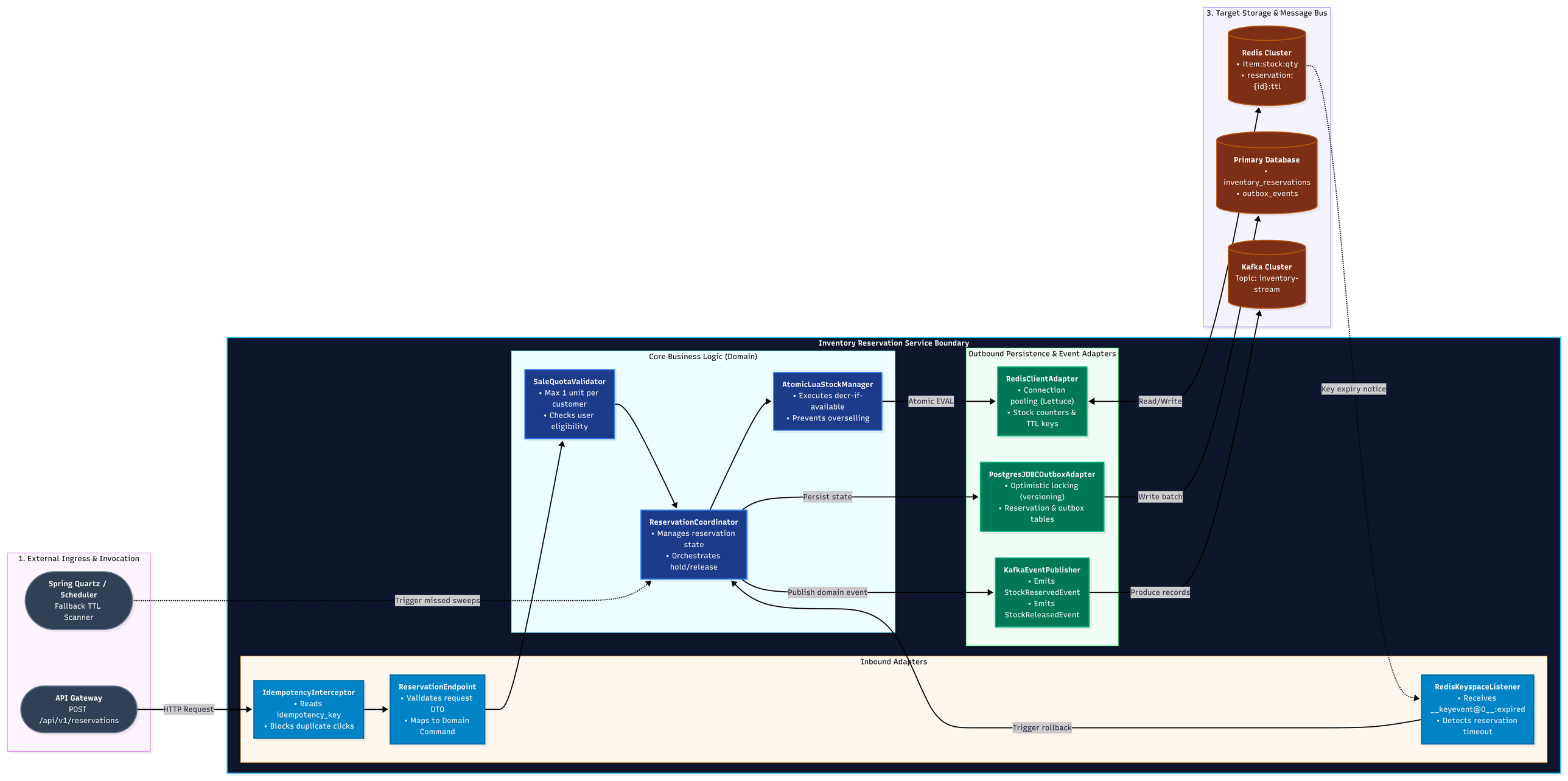

1. External Ingress & Invocation
Spring Quartz / Scheduler
Fallback TTL Scanner
API Gateway
POST /api/v1/reservations
HTTP Request
Trigger missed sweeps
Inventory Reservation Service Boundary
Inbound Adapters
IdempotencyInterceptor
▪ Reads idempotency_key
▪ Blocks duplicate clicks
ReservationEndpoint
▪ Validates request DTO
▪ Maps to Domain Command
Core Business Logic (Domain)
SaleQuotaValidator
▪ Max 1 unit per customer
▪ Checks user eligibility
AtomicLuaStockManager
▪ Executes decr-if-available
▪ Prevents overselling
ReservationCoordinator
▪ Manages reservation state
▪ Orchestrates hold/release
Atomic EVAL
Persist state
Publish domain event
Outbound Persistence & Event Adapters
RedisClientAdapter
▪ Connection pooling (Lettuce)
▪ Stock counters & TTL keys
PostgresJDBCOutboxAdapter
▪ Optimistic locking (versioning)
▪ Reservation & outbox tables
KafkaEventPublisher
▪ Emits StockReservedEvent
▪ Emits StockReleasedEvent
Read/Write
Write batch
Produce records
3. Target Storage & Message Bus
Redis Cluster
▪ item:stock:qty
▪ reservation:{id}:ttl
Primary Database
▪ inventory_reservations
▪ outbox_events
Kafka Cluster
Topic: inventory-stream
Key expiry notice
RedisKeyspaceListener
▪ Receives __keyevent@0__:expired
▪ Detects reservation timeout
Trigger rollback